

Theory of Computation

Exercise 6: (Pumping Lemma)

Prove by Pumping Lemma that the following languages are not regular.

1. $L1 = \{ 0^i 1^j : j \text{ is a multiple of } i \}$

The whiteboard contains the following content:

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Exercise 6:
(Pumping Lemma)

Prove by Pumping Lemma that the following languages are not regular.

1. $L1 = \{ 0^i 1^j : j \text{ is a multiple of } i \}$

- Assume that $L1$ is regular. Let m be the number of states in DFA.
- We pick a string $w \in L1$, $|w| \geq m \Rightarrow w = 0^m 1^m$
- Opponent decomposes w into xyz , $|xy| \leq m$, $|y| \geq 1$
 $y = 0^k$
Diagram: $\overbrace{0000}^m \cdot \overbrace{1111}^m$
 $\underbrace{\hspace{1cm}}_{x+y} \quad \underbrace{\hspace{1cm}}_z$
- We pick $i=2$ to pump xy^iz . The result is $0^{m+k} 1^m$ where $k=|y|$ and $k \geq 1$.
- Since m is not a multiple of $m+k$, $0^{m+k} 1^m \notin L1$.
- Contradiction with Pumping Lemma! Therefore, $L1$ is not regular language.

2. $L_2 = \{ 0^i 1^j : i < j \}$

2. $L_2 = \{ 0^i 1^j : i < j \}$

- Assume that L_2 is regular. Let m be the number of states in DFA.

- We pick a string $w \in L_2$, $|w| \geq m \Rightarrow w = 0^m 1^{m+1}$

- Opponent decomposes w into xyz , $|xy| \leq m$, $|y| \geq 1$

$\overbrace{0000\dots 0}^m \overbrace{111\dots 1}^{m+1}$
 $\underbrace{\hspace{1.5cm}}_{x+y} \quad \underbrace{\hspace{1.5cm}}_z$

- We pick $i=2$ to pump xy^iz . The result is $0^{m+k}1^{m+1}$ where $k=|y|$ and $k \geq 1$.

- Since $m+k \geq m+1$, $0^{m+k}1^{m+1} \notin L_2$.

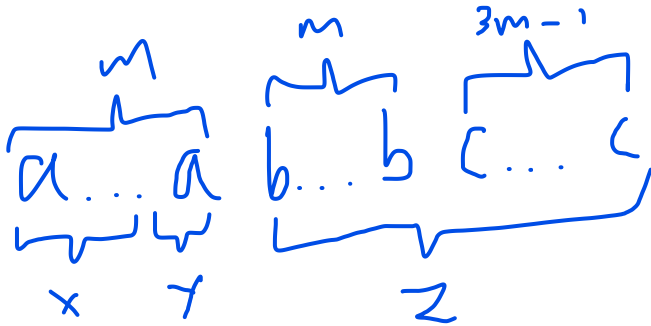
- Contradiction with Pumping Lemma! Therefore, L_2 is not regular language.

*3. Prove that the following language is not Regular

$$L = \{ a^i b^j c^k : k < 2i + j \}$$

- Assume that L is regular. Let m be the number of states in DFA
- We pick a string w in L, $|w| \geq m \rightarrow w = a^m b^m c^{3m-1}$
- Opponent decompose w into xyz, $|xy| \leq m$, $|y| \geq 1$

$$a^m \quad b^m \quad c^{3m-1}$$



- Let $y = a^k$, $|y| = k$
- we choose $i = 0$ to pump $xy^i z$. The result is
- Since $2(m-k) + m < 3m-1$, $k \geq 1$ not in L

$$a^{m-k} \quad b^m \quad c^{3m-1}$$

$$2m - 2k + m < 3m - 1 \quad ; k \geq 1$$

$$3m - 2k < 3m - 1 \quad ; k \geq 1$$

- Therefore L is not regular